We know that any surface (smooth or rough) can absorb sound. For example, the sound produced in a big hall or auditorium or theatre is absorbed by the walls, ceilings, floor, seats etc. To avoid such losses, a curved sound board (concave board) is kept in front of the speaker, so that the board reflects the sound waves of the speaker towards the audience. This method will minimize the spreading of sound waves in all possible directions in that hall and also enhances the uniform distribution of sound throughout the hall. That is why a person sitting at any position in that hall can hear the sound without any disturbance.

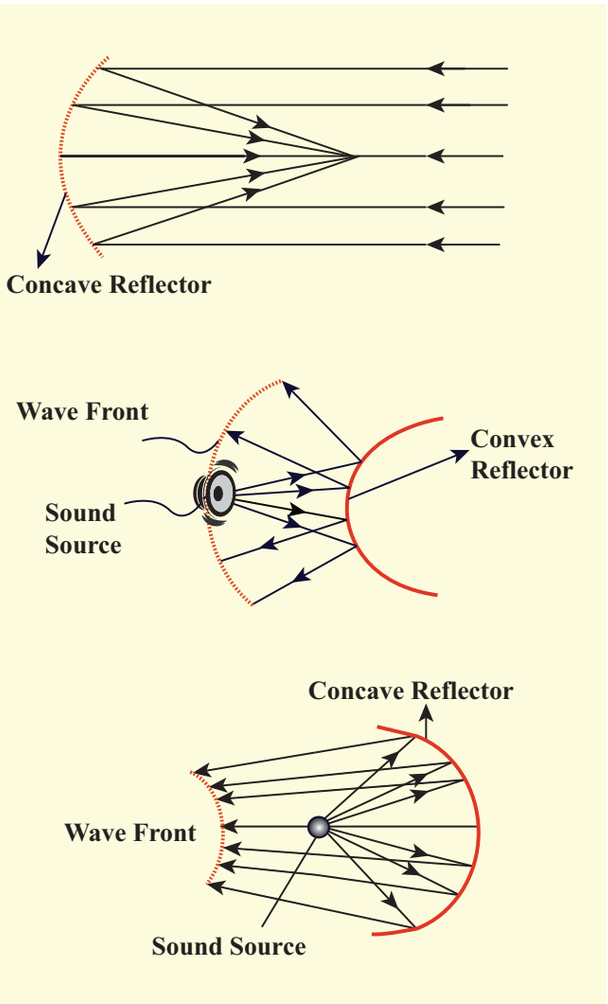


Figure 11.20 Reflection of sound through the curved surface

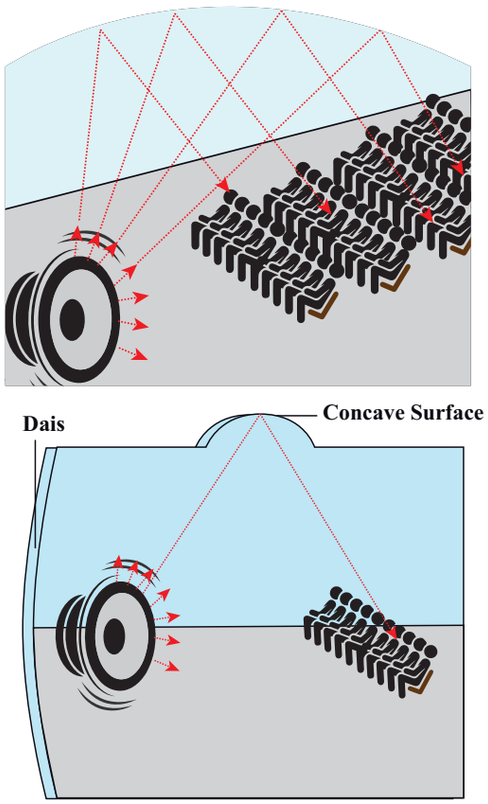


Figure 11.21 Sound in a big auditorium

11.5.3 Applications of reflection of sound waves

(a) Stethoscope: It works on the principle of multiple reflections.

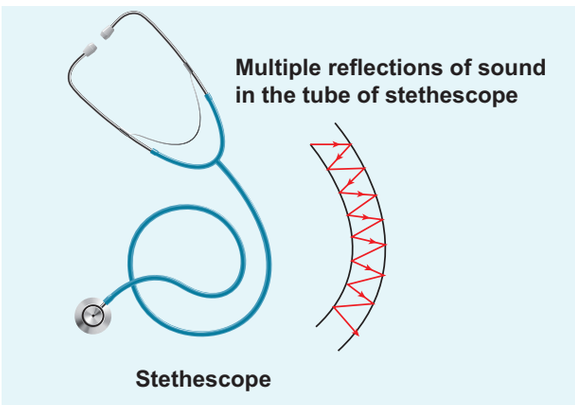


Figure 11.22 Stethoscope and multiple reflection of signal in a rubber tube

UNIT-11(XI-Physics_Vol-2)_14-08-2020.indd 241 17-08-2020 15:30:54